ROADWATCH

Requirements Specification | Project Assignment #2

University of Nevada, Reno

Department: Computer Science & Engineering

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Team Number: 19

Team Name: RoadWatch

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2. Introduction

RoadWatch is a pothole detection and monitoring system designed to automate part of the roadway inspection process. The system will process recorded road video, use computer vision to identify potholes, associate detections with location and time data, store the results in a database, and display detected potholes on an interactive map. The goal is to reduce reliance on manual observation while giving maintenance personnel organized information that can support roadway inspection and repair planning.

Since PA-1, the overall purpose of RoadWatch has remained the same, but several aspects of the project have become more clearly defined. Feedback from Travis Truhill, Director of Maintenance and Operations for the City of Reno, helped clarify how a system like RoadWatch could fit into an actual municipal maintenance workflow. Precise geolocation is especially important, along with the ability to view and group nearby potholes on a map. Status tracking was also identified as useful because the City currently manages road maintenance through work orders that move through open, pending, and completed stages. In addition, pothole severity was identified as a valuable feature for prioritizing repairs, although accurately estimating severity from ordinary camera footage may be difficult because of shadows, debris, and varying road conditions.

These findings have helped refine the project scope and priorities. The core system will continue to focus on reliable pothole detection, GPS association, database storage, and map visualization. More advanced capabilities, such as severity classification, geographic clustering, detailed roadway-position information, and integration with municipal work-order systems, may be treated as later-priority features depending on technical feasibility and available development time. The team has also clarified that the current version of RoadWatch will process uploaded recorded footage offline rather than performing computer-vision processing directly on the mobile recording device. This creates a clearer separation between data collection and processing and better defines the architecture for the current project scope.

The team also has a clearer understanding of how RoadWatch could eventually be deployed. Rather than requiring dedicated inspection vehicles, the system could potentially process footage collected by cameras on vehicles already traveling city streets, allowing roadway conditions to be monitored with less manual effort.

3. Stakeholder Interviews

3.1 Stakeholders

Travis Truhill - Director of Maintenance and Operations, City of Reno

Travis Truhill was selected because he provides expertise related to roadway maintenance, pothole reporting, repair prioritization, and municipal maintenance workflows. His role gives us insight into how potholes are currently identified, evaluated, tracked, and repaired by the City of Reno. His feedback is especially relevant for determining which RoadWatch features would provide practical value to a real public-works organization.

John Thrasher - Data 360 Architect

John Thrasher was selected because he provides a technical perspective on data architecture, database design, system integration, reliability, and scalability. His experience in the computer science field and current role as a Data 360 Architect make his feedback especially useful for determining how RoadWatch should structure, validate, store, and manage pothole data. His input will also help identify important non-functional requirements related to maintainability, security, scalability, and data integrity.

Victor Rosales - Proxy User

Victor Rosales was selected to act as a proxy user and provide feedback on RoadWatch's usability and overall user experience. His perspective helps identify how simple the video upload process should be, what information users would expect to see on the map and detection screens, and how the system should handle incorrect detections or processing errors.

3.2 Interview Questions and Findings

Question 1

Stakeholder: Travis Truhill

Question:
How does the City of Reno currently identify, report, and track potholes, and what are the biggest challenges with that process?

Summary of Response:
The City of Reno currently relies mainly on reports from citizens and observations from city employees working in the field. Citizens can report potholes through Reno Direct, while city staff can report roadway problems through their supervisors. Reno Direct uses ServiceNow for

customer-service management, and ServiceNow communicates with the City's Elements asset-management system to create work orders for maintenance crews.

Travis explained that the current identification process is still largely manual and that limited funding has prevented the City from implementing more advanced roadway-monitoring technology. The most time-consuming part of the process is evaluating reported potholes to determine whether they create a serious public-safety issue. The City now prioritizes severe hazards and groups lower-priority potholes geographically so crews can repair several in one area instead of traveling across the city for individual requests.

Key Insight / Effect on Requirements:
RoadWatch should reduce reliance on manual observation and reporting by automatically identifying potholes. The system should also support geographic grouping and provide enough information to help maintenance personnel evaluate and prioritize detections.

Question 2

Stakeholder: Travis Truhill

Question:
If RoadWatch automatically detected potholes from road video, what information would be most important to store for each detection?

Summary of Response:
Travis identified precise geographic location as the most important information. Latitude and longitude are essential, but he noted that the system would be even more useful if it could identify the pothole's exact roadway position, including direction of travel and lane location.

He also emphasized the usefulness of mapping detections visually. A system that could place potholes on a map and show several detections within the same area would help maintenance supervisors allocate their crews more efficiently.

Key Insight / Effect on Requirements:
Each pothole detection should store precise GPS information and be displayed on an interactive map. Geographic clustering or grouping should also be considered because it directly supports the City's current maintenance workflow.

Question 3

Stakeholder: Travis Truhill

Question:
How useful would features like pothole severity, duplicate detection, repair status, and filtering be, and which of those would you prioritize first?

Summary of Response:
Travis identified pothole severity as the most important of the proposed features. Severity is important because the City already prioritizes potholes based on the level of public-safety risk they present.

He also identified a technical concern with automatically determining severity from camera footage. Shadows, debris, lighting conditions, and shallow pavement delamination could make it difficult for a camera-based system to accurately determine the depth or seriousness of a pothole.

Key Insight / Effect on Requirements:
Severity classification would provide substantial value to maintenance personnel, but it may be technically difficult to implement reliably. Because of this, severity should be considered a high-value feature but may be more appropriate as a later-priority requirement if it cannot be completed reliably during the initial project phase.

Question 4

Stakeholder: Travis Truhill

Question:
Would collecting road video through phones or cameras on city vehicles be practical, and is there any existing pothole, GIS, maintenance, or road-condition data that could potentially help us test the system?

Summary of Response:
The City currently uses Samsara camera systems that capture video and still images. Travis was unsure whether that data could be shared because the camera platform may be proprietary.

He described a previous system in which street-sweeper operators could manually press a button whenever they observed a roadway problem. The system would record the location, but the process became unreliable because workers did not consistently use the button. Travis explained that an automated system would solve this problem by removing the need for employees to manually identify and record each issue.

Street sweepers may be particularly useful for future RoadWatch deployment because they regularly travel city streets. Travis also offered to show the team the City's asset-management, GIS, and vehicle workflows in person.

Key Insight / Effect on Requirements:
RoadWatch should be designed so that pothole detection can occur automatically from recorded roadway footage without requiring manual input from drivers. Using cameras mounted on vehicles that already travel city streets represents a realistic future deployment model.

Question 5

Stakeholder: Travis Truhill

Question:
How does the City decide which potholes should be repaired first, and would RoadWatch be more useful if it helped support that prioritization?

Summary of Response:
The City prioritizes potholes based primarily on public-safety risk. For example, a pothole located in the wheel path of an active travel lane would receive higher priority than a minor surface defect or damage located near the shoulder.

Critical hazards may be repaired immediately, while less urgent potholes are grouped geographically. Maintenance crews are then sent to a specific area to repair several potholes during the same trip. Travis explained that this approach is much more efficient than responding to potholes strictly according to how long ago they were reported.

Key Insight / Effect on Requirements:
RoadWatch would be more useful if it supported both pothole prioritization and geographic grouping. Severity, roadway position, and clustering could eventually help maintenance supervisors determine which detections require immediate attention and which can be grouped into scheduled work.

Question 6

Stakeholder: Travis Truhill

Question:
Once a pothole is repaired, how is that tracked, and would statuses such as detected, verified, scheduled, and repaired be useful in the system?

Summary of Response:
Travis said that status tracking would be very useful. The City currently manages pothole work through work orders in the Elements asset-management platform. Work orders can be marked as open, pending, or closed/completed.

An open work order has not yet been addressed. A pending work order has generally been evaluated and may be grouped with other work that needs to be completed. Once the repair is finished, the work order is closed and can include repair notes.

The City can query this information to track outstanding work and generate statistics about the number of requests received, completed work, and overall performance.

Key Insight / Effect on Requirements:
RoadWatch should support status tracking for detections and should store data in a way that allows users to query and report on pothole records. A workflow such as detected, verified, pending/scheduled, and repaired could align well with existing municipal maintenance processes.

Question 7

Stakeholder: John Thrasher

Question: What data should RoadWatch require for every pothole record, and how should that data be structured so it is easy to store, search, update, and display?

Summary of Response:
John recommended storing location, GPS accuracy, timestamp, confidence score, an image reference, and the version of the computer-vision model that produced each detection. He suggested separating individual detections from confirmed potholes because the same pothole may be detected multiple times on different drives. A detection should therefore be allowed to remain unlinked until it is matched to an existing pothole or confirmed as a new pothole.

He also recommended using spatial database capabilities such as PostGIS for location data and storing images outside the database, with only their URLs stored in the pothole record. Status values should use predefined options rather than unrestricted text. He also noted that vague fields such as “characteristics” should be replaced with clearly defined database fields.

Key Insight / Effect on Requirements:
RoadWatch should distinguish between a raw pothole detection and a confirmed pothole record rather than treating every detection as a completely new pothole. Detection records should include clearly defined metadata such as GPS accuracy and model version, while images should be referenced rather than stored directly in the database. The database design should also support spatial queries and standardized status values.

Question 8

Stakeholder: John Thrasher

Question: How should RoadWatch handle incomplete, invalid, or duplicate pothole data so that the database stays accurate and reliable?

Summary of Response:
John recommended validating every detection before it is accepted into the main database. Records with clearly invalid information, such as missing GPS coordinates or confidence scores outside the expected range, should be rejected or separated from valid records. Rejected detections should still be recorded with the reason for rejection so recurring computer-vision or data-collection problems can be identified.

Detections with poor GPS accuracy or low model confidence should be flagged for review rather than immediately displayed as valid potholes. Unique identifiers should be used to prevent duplicate records caused by repeated uploads or retries. Because GPS locations may vary between drives, duplicate pothole matching should use an approximate geographic radius and allow a reviewer to manually merge or separate records when necessary.

Key Insight / Effect on Requirements:
RoadWatch needs explicit data-validation and duplicate-handling rules. Invalid or questionable detections should be flagged or separated instead of being treated as confirmed potholes, and the system should support human review when automated duplicate matching is uncertain. This strengthens the need for detection verification and manual merge/split functionality.

Question 9

Stakeholder: John Thrasher

Question: How should the computer-vision component, backend/database, and web dashboard communicate with each other to keep the system maintainable and easy to expand?

Summary of Response:
John recommended placing a single backend API between the computer-vision system, database, and web dashboard. This would prevent individual components from depending directly on the database structure and would make future changes easier to manage.

Video processing should happen through a background job queue because computer-vision processing may take longer than normal web requests. The team should define and document the JSON data exchanged through the API so different parts of the project can be developed independently. He also recommended including a classification field in the data model from the beginning so additional road-damage types can be supported later without redesigning the system. For a four-person project team, he recommended a modular monolithic architecture rather than the added complexity of microservices.

Key Insight / Effect on Requirements:
RoadWatch should use a clearly separated architecture consisting of the computer-vision component, backend API/database layer, and web interface. Video-processing tasks should run independently from normal web requests, and the communication format between components should be defined in advance. This supports maintainability and future expansion without introducing unnecessary architectural complexity.

Question 10

Stakeholder: John Thrasher

Question: What reliability, scalability, security or privacy concerns should we consider when storing road video, GPS data, images, and pothole records?

Summary of Response:
John identified privacy and storage concerns as especially important if RoadWatch requires users to upload full roadway videos. Raw footage may contain license plates, pedestrians, homes, and other potentially identifying information, so he recommended deleting raw videos after processing and retaining only the pothole images needed by the system. GPS tracks could also have their beginning and ending portions removed when appropriate to reduce the risk of exposing private locations.

For security, he recommended role-based access, encrypted HTTPS communication, private image/video storage, temporary access links, file validation, and rate limiting. Human review should occur before questionable detections become publicly visible.

For reliability, uploads should support automatic retries without creating duplicate records, and the database should be backed up regularly with recovery procedures tested. For scalability, he identified video storage and computer-vision processing as larger concerns than database storage itself. Spatial indexing should be used so the map only retrieves potholes relevant to the area currently being viewed.

Key Insight / Effect on Requirements:
RoadWatch should minimize the amount of raw video it permanently stores and protect both uploaded media and location data. Security, backup/recovery, safe upload handling, and duplicate prevention should be explicitly addressed by the system design. Spatial database indexing and selective map loading should also be considered to keep the system responsive as the number of detections grows.

Question 11

Stakeholder: Victor Rosales

Question: If you were using RoadWatch for the first time, what would make the process of uploading road video and getting results feel simple and intuitive?

Summary of Response:

If it was my first time using RoadWatch I believe that the process of uploading videos of the road would be feel simple and intuitive if we were able to do the bare minimum. I would not want to have to keep touching the phone recording the road, I would like it if everything was done automatically and once the capturing was done then maybe I hit a send button and possibly fill out a tiny description of the area that I was in but the less I have to do the better.

Key Insight / Effect on Requirements:

The system needs to be as hands off as possible, we want to preserve as much of the workers time as possible. In doing so we need an easy to set up system where the device is set up at an angle in which it can easily detect the potholes and not make it difficult for the worker to send out the footage that was captured.

Question 12

Stakeholder: Victor Rosales

Question: When viewing detected potholes, what information would you want visible on the map, and what additional details would you expect after clicking a pothole?

Summary of Response:

I would like to see the physical location itself on the map so maybe like a red dot as an indicator of where on the road the pothole is. I would also like to see either a deepened red color or some sort of indicator to tell me the severity of each pothole. Also maybe something that groups the potholes in a mini window on the side of the screen that shows the potholes in the area depending on how zoomed in I am on the map.

Key Insight / Effect on Requirements:

We need to make sure that our visual representation of the potholes have some sort of way to determine quickly the severity of the pothole without having to click into it. I believe that having a color system where depending on the severity of the pothole, we visualize it using a deeper color. For example a yellow dot may represent a pothole with a severity between 50-70%, an orange dot means a severity of 70-80% and finally a red color in which we can have a light red color which represents about 85-95% and a dark red which represents a very severe pothole which poses immediate danger to the public (>95%).

Question 13

Stakeholder: Victor Rosales

Question: If RoadWatch made an incorrect detection or something went wrong during processing, what would you expect the system to let you do or tell you?

Summary of Response:

I would expect an error message that would tell me exactly what the problem is and possibly a tip on how to fix it underneath so I could troubleshoot the issue right away before I reach out for help. I would also like to keep a photo of the pothole itself just in case the system is unsure if it has detected a pothole and this would need a manual review which is the way the city currently does it anyway. I would hope that errors would be minimal and that way the manual evaluation process is sped up.

Key Insight / Effect on Requirements:

Roadwatch should implement an error detection system which lets the user know when something has malfunctioned or if something may be wrong. We also need to keep at least an image of the “detected pothole” just in case a manual review is needed.

4. Technical Requirements Specification

RoadWatch requirements are prioritized across three levels based on importance and expected implementation timeline. Level 1 requirements define the core functionality that must be implemented by the end of Fall 2026. Level 2 requirements are targeted for Spring 2027, while Level 3 requirements represent future capabilities that may be implemented beyond Spring 2027.

4.1 Functional Requirements

4.1.1 Level 1 - Must Have

Req ID	Requirement Statement	Acceptance Criteria	Verification Method
FR-1	RoadWatch must allow recorded roadway video to be provided to the	A supported video file can be submitted to the system and successfully passed to	Integration test & live demo

	system for processing.	the pothole-detection component without failure.	
FR-2	RoadWatch must analyze recorded roadway video and identify potholes using computer vision.	A labeled validation video containing known potholes can be processed, and the system outputs detection locations and confidence scores that can be evaluated against the manually labeled ground truth using precision and recall.	Model evaluation & live demo
FR-3	RoadWatch must associate each pothole detection with geographic location data.	Every stored pothole detection has valid latitude and longitude values when corresponding GPS data is available.	Integration test & database inspection
FR-4	RoadWatch must associate each pothole detection with the time at which it occurs in the recorded footage.	Every stored detection contains a timestamp corresponding to its location in the processed video.	Integration test & database inspection
FR-5	RoadWatch must store pothole detections in a database.	Each successfully processed detection creates a database record containing a unique identifier, GPS coordinates, GPS accuracy when available, timestamp, image reference, confidence score, and model version.	Database test & integration test
FR-6	RoadWatch must save an image or video frame representing each detected	Each stored detection can be associated and retrieve an image or frame showing the	Integration test & live demo

	pothole.	detected pothole.	
FR-7	RoadWatch must store the computer-vision model's confidence score for each pothole detection.	Every detection record has a valid confidence value produced by the detection model.	Database inspection & automated test
FR-8	RoadWatch must show stored pothole detections on an interactive web map.	Stored detections with valid coordinates appear as markers at their corresponding locations on the map.	Integration test & live UI demo
FR-9	RoadWatch must allow a user to see information about an individual pothole detection.	Selecting a pothole marker shows its available location, timestamp, image, and confidence information.	UI test & live demo

4.1.2 Level 2 - Should Have

FR-10: RoadWatch must classify detected potholes by severity to help users distinguish higher-priority roadway hazards from less serious defects.

FR-11: RoadWatch must allow nearby pothole detections to be geographically grouped or clustered to support efficient maintenance planning.

FR-12: RoadWatch must allow users to filter or search stored detections using available information like location, date, confidence, severity, or status.

FR-13: RoadWatch must allow a user to verify, reject, or review automatically generated pothole detections.

FR-14: RoadWatch must support status tracking for pothole detections, such as detected, verified, scheduled, and repaired.

FR-15: RoadWatch must give basic statistics or reports from stored pothole data, like total detections, detections by status, and detections within a selected area.

FR-16: RoadWatch must associate additional roadway-position information, like direction of travel or lane, with a detection when that information can be determined.

FR-17: RoadWatch must identify detections that may represent the same pothole and allow a reviewer to merge or separate those records when necessary.

4.1.3 Level 3 - Nice to Have

FR-18: RoadWatch must support integration with external municipal asset-management or work-order systems.

FR-19: RoadWatch must support automatic creation of maintenance work orders from reviewed or verified pothole detections.

FR-20: RoadWatch must support direct or continuous collection of roadway footage from cameras installed on municipal fleet vehicles.

FR-21: RoadWatch must support detection and classification of additional roadway-damage types beyond potholes.

FR-22: RoadWatch must support historical tracking of roadway defects to help monitor deterioration or changes over time.

FR-23: RoadWatch must use historical pothole, roadway, and environmental data to identify areas that may be at higher risk of developing potholes.

FR-24: RoadWatch must use predictive machine learning to estimate future pothole risk based on factors like weather conditions, location, and previously detected roadway damage.

FR-25: RoadWatch must display predicted high-risk roadway areas on the map to help users plan preventive inspections or maintenance.

4.2 Non-Functional Requirements

NFR-1 - Performance: After detection data has been retrieved from the database, the web interface must display the requested pothole information within 2 seconds under normal test conditions.

NFR-2 - Reliability: Successfully stored pothole records must remain available after the RoadWatch app is restarted.

NFR-3 - Data Integrity: RoadWatch must validate detection data before it is accepted into the main dataset. Records missing required fields or containing invalid values must be rejected or flagged for review.

NFR-4 - Usability: A first-time user must be able to open the RoadWatch dashboard, view pothole markers, select a detection, and see its associated information without developer assistance.

NFR-5 - Maintainability: The computer-vision, backend/database, and web-interface components must be organized as separate components so that one component can be modified or replaced without needing the entire system to be rewritten.

NFR-6 - Security: RoadWatch must restrict database modification to authorized application/backend interfaces, use HTTPS for network communication, and prevent unrestricted public access to stored media.

NFR-7 Scalability: RoadWatch must be designed so that increasing numbers of pothole detections can be stored and retrieved without requiring changes to the database schema.

NFR-8 - Privacy: Raw roadway video must not be retained permanently after successful processing unless needed for testing or authorized system evaluation. RoadWatch should retain only the detection images and metadata required for system operation. \

NFR-9 - Recovery: RoadWatch must maintain backups of persistent pothole records so stored detection information can be recovered following a database failure.

4.3 Hardware Requirements

RoadWatch does not require specialized proprietary hardware for the current MVP. The system requires roadway footage recorded by a smartphone. GPS information must be available from the recording device itself. Development and processing will be performed using standard personal computers capable of running the RoadWatch computer-vision, backend, database, and web components. Future versions of RoadWatch may integrate directly with cameras or GPS systems installed on municipal vehicles, but such integration is outside the current level 1 scope.

5. Use Case Modeling

5.1 Use Case Diagram

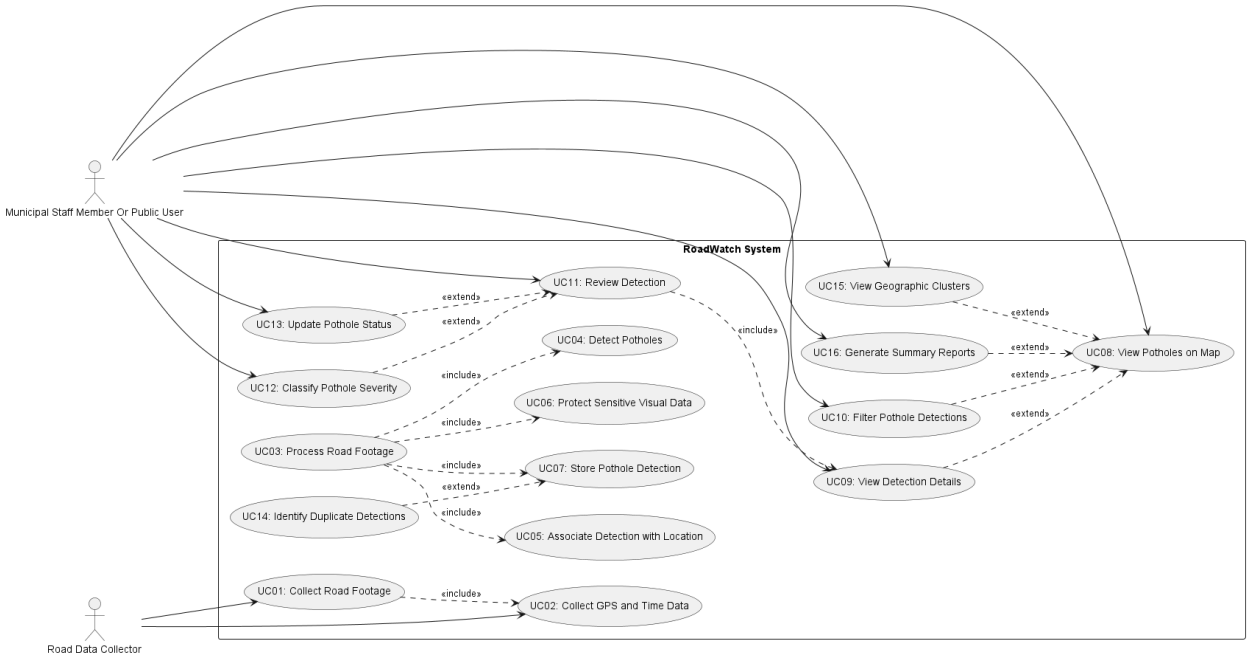


Figure 5.1.1: Use case diagram for RoadWatch, showing data collection, automated processing, and municipal dashboard functionality. Include and extend represent relationships between Use Cases.

5.2 Use Case Summaries

ID	Use Case	Description
UC01	Collect Road Footage	Data collection utilizes a smartphone to record roadway footage for pothole detection. The recording workflow is designed to minimize driver interaction and distraction.
UC02	Collect GPS and Time Data	RoadWatch collects geographic coordinates and timestamps corresponding to the recorded footage. This information provides the location and time context required for pothole detections.
UC03	Process Road Footage	RoadWatch processes the recorded footage through its computer-vision pipeline and generates pothole detection records.
UC04	Detect Potholes	The computer-vision model analyzes road imagery to identify potential potholes and produces detection results,

		including confidence scores.
UC05	Associate Detection with Location	Each detected pothole gets associated with the corresponding GPS coordinates and timestamp when valid location data is available.
UC06	Protect Sensitive Visual Data	Privacy safeguards are applied to prevent unnecessary retention or transmission of sensitive visual information, such as identifiable pedestrians and license plates.
UC07	Store Pothole Detection	Valid pothole detection records and their associated metadata are stored in the database for subsequent retrieval and visualization.
UC08	View Potholes on Map	Municipal staff access an interactive map displaying stored pothole detections at their geographic locations. A public version will also be available.
UC09	View Detection Details	Municipal staff select an individual detection to inspect its image, location, timestamp, confidence score, and other available metadata.
UC10	Filter Pothole Detections	Municipal staff filter the map's detections using available criteria, such as location, collection date, severity, or review status.
UC11	Review Detection	Municipal staff inspect a potential pothole detection and confirm or reject it when manual verification is supported.
UC12	Classify Pothole Severity	Severity category is assigned to a pothole using the project's defined classification approach, helping staff assess the relative importance of detections.
UC13	Update Pothole Status	Municipal staff update a detection's maintenance status, such as verified, scheduled for repair, or repaired, if status management is included in the project scope.
UC14	Identify Duplicate Detections	Potentially repeated detections of the same physical pothole are identified so they can be reviewed and handled.
UC15	View Geographic Clusters	Municipal staff inspect groups of nearby detections to identify areas with multiple recorded potholes.
UC16	Generate Summary Reports	Municipal staff view summary statistics about recorded potholes, such as detection counts by location, severity, or status.

Table 5.2.1: Detailed use case descriptions for RoadWatch.

5.3 Detailed Use Case Templates

Use Case: View Potholes on Map	
Use Case ID (UCID)	UC08
Actors	Municipal Staff Members and Public User
Precondition(s)	1. The web dashboard is accessible. 2. The database contains stored pothole detections, or the system can display an empty map. 3. The map service and detection data service are operational.
Flow of Events	1. The municipal staff or public user opens the dashboard. 2. The dashboard requests stored pothole detections from the database. 3. Retrieves detections with valid geographic coordinates. 4. The dashboard displays the retrieved detections as markers on an interactive map. 5. The staff member or public user pans or zooms the map to inspect an area of interest. 6. The staff member or public user selects a marker to view the corresponding detection details.
Alternative / Expectation Flows	1. If no detections are available, the map is displayed with an appropriate empty-state message. 2. Detections without valid coordinates are not placed at geographic locations. 3. If data retrieval fails, an error is displayed and does not misrepresent incomplete results as a successful load.
Postcondition(s)	1. Done when every retrieved detection with valid coordinates is represented by a marker at its corresponding location. 2. Done when selecting a marker allows the user to access the correct detection record. 3. Done when the map provides understandable feedback when no detections are available or retrieval fails.

Table 5.3.1: Template for View Potholes on Map use case.

Use Case: Collect Road Footage	
Use Case ID (UCID)	UC01
Actors	Road Data Collector
Precondition(s)	1. The data-collection component is installed or accessible on a supported smartphone. 2. The smartphone has access to its camera and location services. 3. The smartphone is securely mounted and the collection process can operate without requiring the driver to interact with the device while driving.
Flow of Events	1. The data collector prepares the smartphone and initiates a road-recording session before driving. 2. RoadWatch begins capturing roadway footage. 3. RoadWatch collects corresponding GPS and timestamp data. 4. RoadWatch records the footage for processing. 5. The recording session ends when the collector stops it or the configured recording conditions are met. 6. RoadWatch makes the collected data available to the processing workflow.
Alternative / Expectation Flows	1. If camera access is unavailable, the user is informed that recording cannot begin. 2. If GPS data is unavailable, RoadWatch records the data-quality issue rather than fabricating coordinates. 3. If storage or recording fails, RoadWatch communicates the failure and does not indicate that a complete recording was saved.
Postcondition(s)	1. Done when a successful recording session produces roadway footage available for processing. 2. Done when corresponding GPS and timestamp data are collected when supported and available. 3. Done when the collection process does not require the driver to interact with the phone while driving.

Table 5.3.2: Template for Collect Road Footage use case.

Use Case: Process Road Footage	
Use Case ID (UCID)	UC03
Actors	RoadWatch Processing System
Precondition(s)	1. Roadway footage has been recorded and made available to the processing pipeline. 2. The computer-vision model and processing components are available. 3. The system has access to the necessary GPS and timestamp data.
Flow of Events	1. RoadWatch receives the recorded roadway data. 2. The system prepares the video or extracted frames for computer-vision analysis. 3. The computer-vision model analyzes the imagery and identifies potential potholes. 4. Each detection is associated with the appropriate timestamp and geographic coordinates when valid data is available. 5. The system generates or saves a representative image and the required detection metadata. 6. RoadWatch applies its privacy safeguards before retaining or transmitting visual data. 7. Valid detection records are stored in the database. 8. The processing workflow reports its outcome and makes successfully stored detections available to the dashboard.
Alternative / Expectation Flows	1. If no potholes are detected, RoadWatch records a successful processing result with zero detections. 2. If GPS data is missing or invalid, the system flags or excludes the location information according to its validation rules. 3. If processing fails, RoadWatch reports the failure and does not mark the entire operation as successful. 4. If a detection record cannot be stored, the system reports the storage error and avoids treating the record as persisted.
Postcondition(s)	1. Done when the processing workflow creates 0+ pothole detections. 2. Done when every successfully stored detection has the required identifier, confidence score, and associated image reference. 3. Done when valid GPS coordinates and timestamps are associated with detections when available. 4. Done when detection records are retrievable by the dashboard. 5. Done when the defined privacy safeguards are applied to transmitted data.

Table 5.3.3: Template for Process Road Footage use case.

Use Case: View Detection Details	
Use Case ID (UCID)	UC09
Actors	Municipal Staff Members and Public User
Precondition(s)	1. The RoadWatch dashboard is accessible. 2. A pothole detection has been stored and is retrievable. 3. The staff member or public user can select the detection from the map or another available results view.
Flow of Events	1. The staff member or public user selects a pothole marker on the map. 2. RoadWatch identifies the selected detection record. 3. The system retrieves the corresponding detection metadata and associated image. 4. The dashboard displays the available location, timestamp, image, confidence score, and other defined characteristics. 5. The staff member or public user reviews the information and returns to the map or selects another detection.
Alternative / Expectation Flows	1. If the image is unavailable, RoadWatch displays the remaining available details and identifies the missing image. 2. If the record cannot be retrieved, the dashboard informs the user that the detection is unavailable. 3. If optional metadata is missing, the dashboard identifies the unavailable information rather than presenting fabricated values.
Postcondition(s)	1. Done when selecting a map marker displays details belonging to the selected detection. 2. Done when the available location, timestamp, image, and confidence score are displayed accurately. 3. Done when missing information and retrieval errors are communicated clearly to the user.

Table 5.3.4: Template for View Detection Details use case.

6. Requirement Traceability Matrix

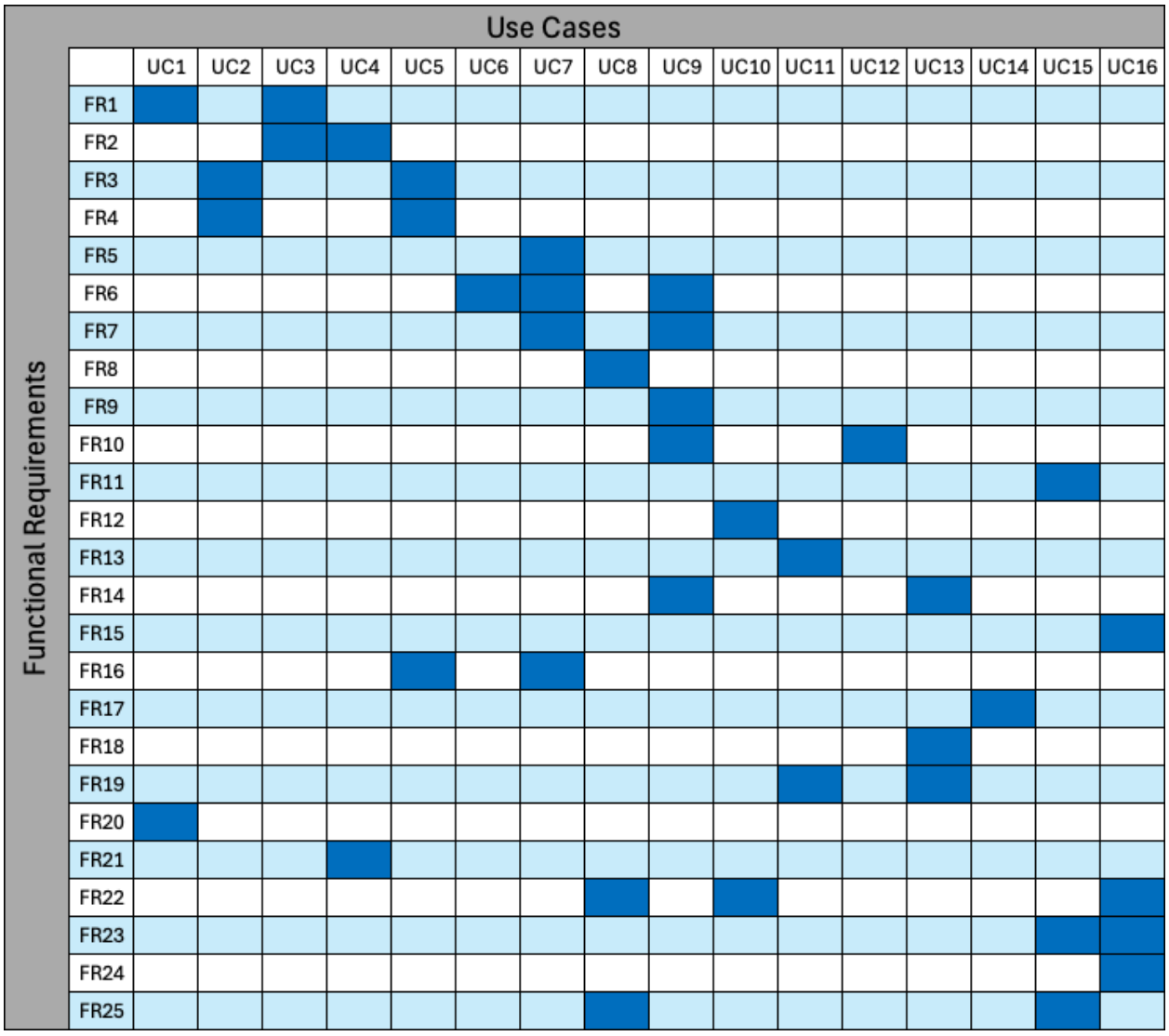


Figure 6.0.1: Requirement Traceability matrix displaying the relationships between the functional requirements and use cases.

7. Initial User Interface Snapshots

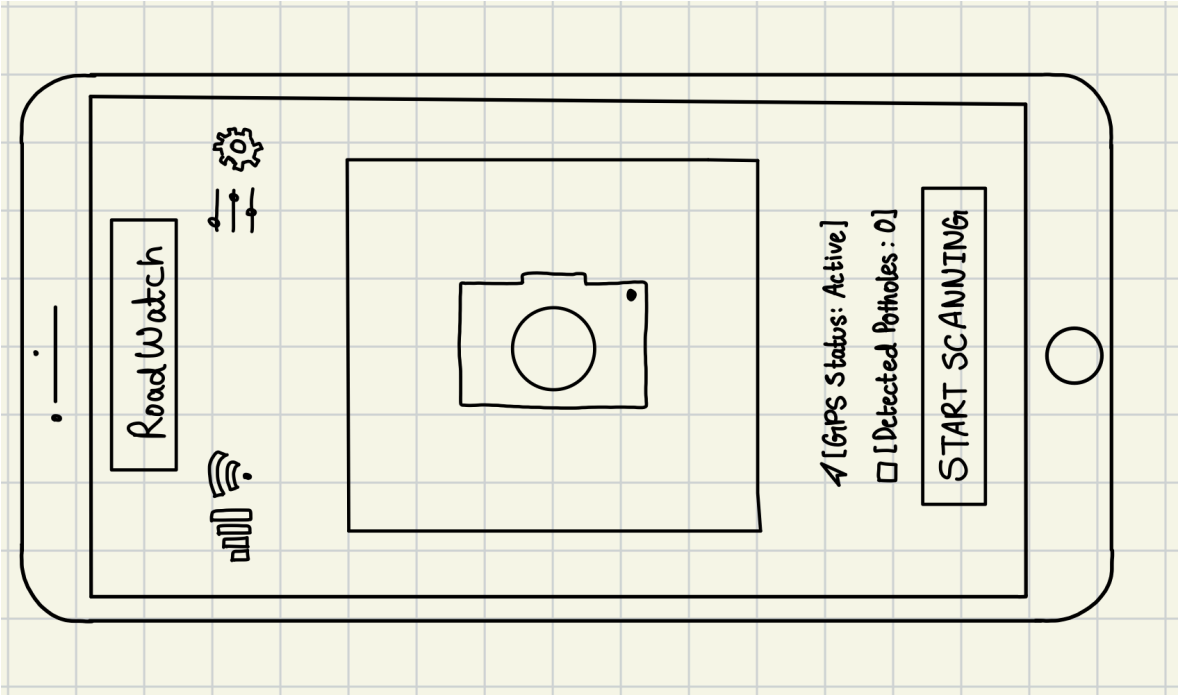


Figure 7.0.1: Potential format of how the application will look like before the user starts to scan/record their drive.

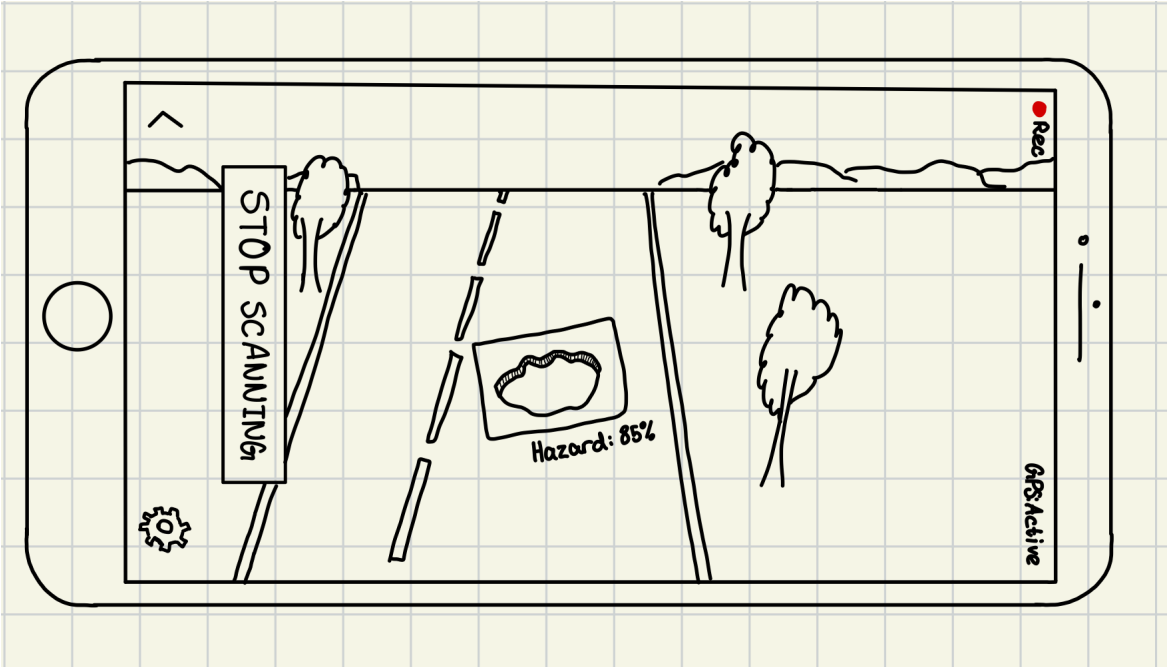


Figure 7.0.2: Potential format of how the app will look like when recording along with a bounding box with the severity of a pothole.

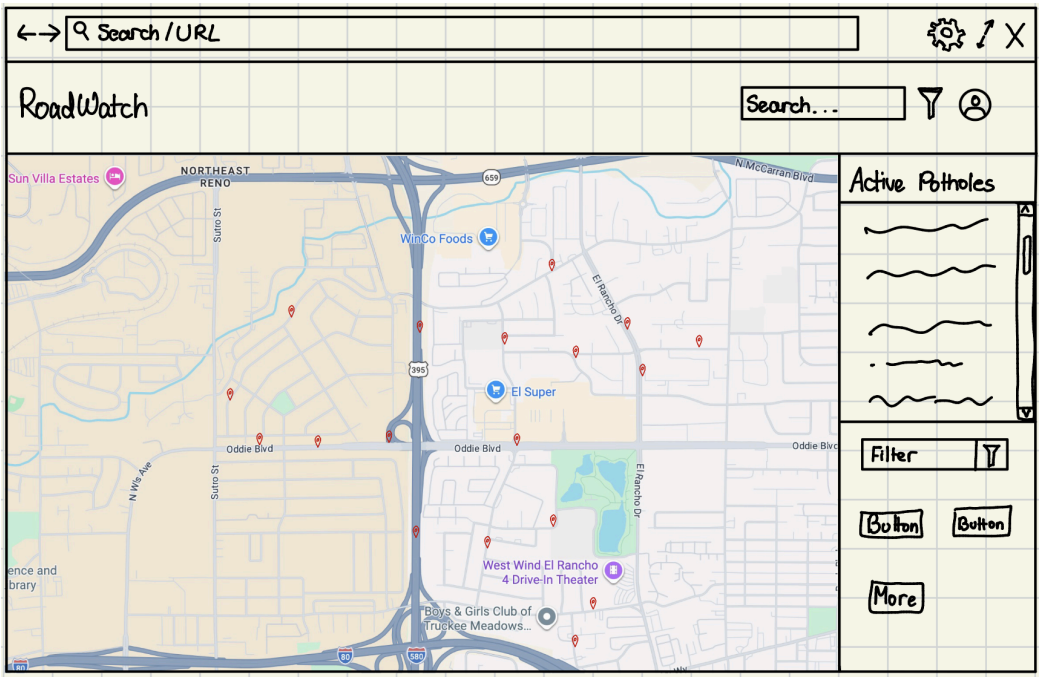


Figure 7.0.3: Potential format of the website on a computer, showing the potholes (red pings) along with active potholes information and other things below it.

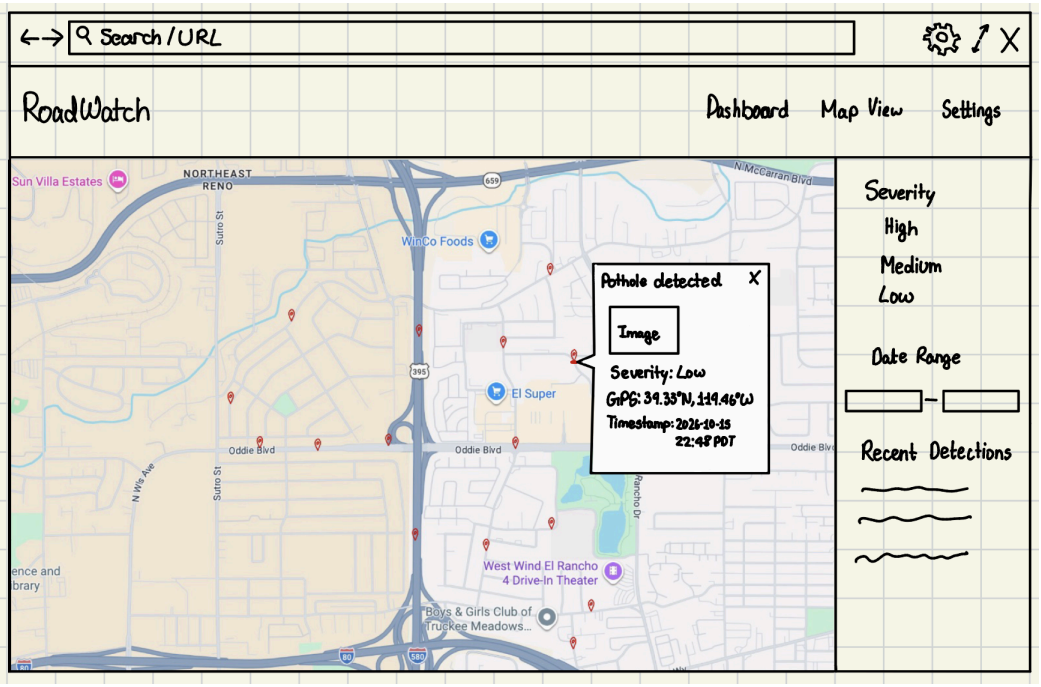


Figure 7.0.4: Potential format of the website after clicking on pothole and getting information about the specific pothole, which has the image of the pothole, the severity, GPS coordinates and the timestamp. More information on right about more potholes in the area and the date range of recorded potholes and recent detections in the area.

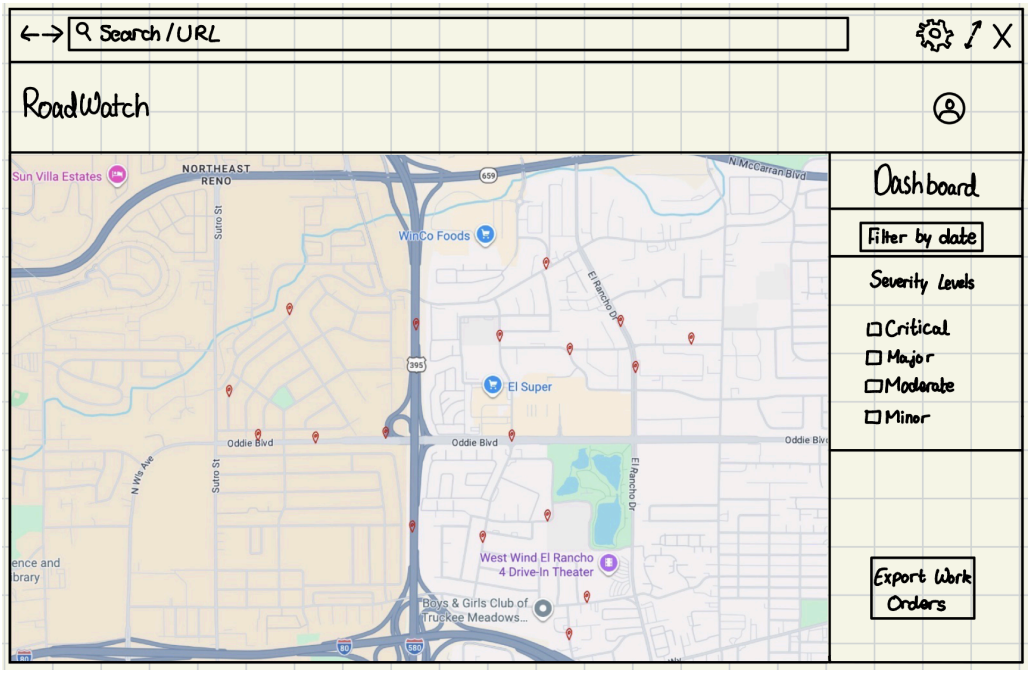


Figure 7.0.5: Potential format of the dashboard which would have an option to filter the potholes by date, and report the potholes based on their severity.

8. Glossary of Terms

Bounding Box: A rectangular border drawn around an object of interest within an image, used by object detection models to indicate the object's severity and size.

Computer Vision (CV): A field of artificial intelligence that enables computers to derive meaningful information from digital images, videos, and visual inputs.

Confidence Score: A percentage value output by a machine learning model indicating how certain it is that object belongs to a specific class.

Fine-Tuning: The process of taking a pre-trained machine learning model and training it further on a specific dataset to improve its accuracy for a specific task.

Object Detection: A computer vision technique that involves identifying and locating instances of objects within an image or video stream.

Pre-trained Model: A machine learning model trained on a large benchmark dataset to solve a problem similar to the one at hand, used as a starting point rather than building a neural network from scratch.

YOLO (You Only Look Once): A highly efficient, real-time object detection algorithm known for its speed and accuracy, acting as the proposed core vision model for the RoadWatch system.

Edge Processing: The practice of processing data locally on the device where it is generated, rather than transmitting raw, heavy video data to a centralized cloud server.

Geo-tagging: The process of adding geographical identification metadata, such as GPS coordinates (Longitude and latitude), to the database logs of detected hazards.

PostgreSQL (Postgres): An open-source relational database management system used as the centralized backend database for storing pothole data, coordinates, and timestamps.

Privacy-by-Design: An engineering approach that integrates privacy protection into the system architecture from the beginning, such as automatically discarding raw video feeds to protect commuter privacy.

Telemetry: The automated collection and transmission of measurements and data (such as GPS location) from remote sources (smartphones) to a central system.

UI/UX (User Interface/User Experience): The design of the visual layout (UI) and the overall interactive flow (UX) that drivers and city workers experience when using the software.

Web Dashboard: The interactive, browser-based graphical interface used by city planners to view, filter, and export the geo-tagged pothole data.

Dashcam: A dashboard camera continuously recording the view through a vehicle's windscreen, representing one of the hardware setups commuters can use for the mobile app.

Passive Scanning: The ability of the mobile application to operate and detect hazards in the background without requiring physical intersection from the driver, ensuring safe operation.

Public Works Department: The local municipal entity responsible for maintaining city infrastructure and roads, acting as the primary target users of the web dashboard.

Severity Rating: An estimated qualitative or quantitative score assigned to a dedicated pothole by the system to help city planners prioritize repairs.

Work Order: A formal request or ticket generated by the city planners for maintenance crews to physically repair a specific infrastructure defect.

RoadWatch: A mobile and web-based infrastructure monitoring system that leverages computer vision and smartphone dashcams to automatically detect, geo-tag, and report road surface hazards to a centralized dashboard for municipal public works departments.

9. References

Book:

Abusamra, Soliman, and El Hakea, Ayman H.. A Comprehensive Guide to Managing Municipal Infrastructure Assets. United States, CRC Press, 2025.

This book examines municipal infrastructure asset management, including data requirements, maintenance planning, performance indicators, and the prioritization of infrastructure investments. It provides background for understanding how municipalities make decisions about roads and other public assets. For RoadWatch, this resource will help our team connect pothole detection and organized condition records to practical maintenance needs, resource allocation, and long-term infrastructure planning.

Peer Reviewed Articles:

Maeda, H., Sekimoto, Y., Seto, T., Kashiyama, T., & Omata, H. (2018). Road damage detection and classification using deep neural networks with smartphone images. *Computer-Aided Civil and Infrastructure Engineering, 33*, 1127–1141. https://doi.org/10.1111/mice.12387

This study investigates automated road damage detection using convolutional neural networks and images captured by a vehicle-mounted smartphone. The researchers created an annotated dataset and evaluated detection accuracy and processing speed on a GPU server and smartphone. This article directly supports RoadWatch's camera-based approach and provides a foundation for selecting detection methods, preparing training data, and evaluating the tradeoff between accuracy and processing speed.

Arya, D., Maeda, H., Ghosh, S. K., Toshniwal, D., & Sekimoto, Y. (2024). RDD2022: A multi-national image dataset for automatic road damage detection. *Geoscience Data Journal, 11*(4), 846–862. https://doi.org/10.1002/gdj3.260

This article introduces RDD2022, a dataset containing 47,420 road images from six countries, including the United States. Its annotations cover longitudinal cracks, transverse cracks, alligator cracks, and potholes. RoadWatch could use this dataset as a starting point for model training and evaluation. Its geographic diversity also highlights the importance of testing our model on local footage before assuming it will perform reliably on Reno roads.

Kim, Y.-M., Kim, Y.-G., Son, S.-Y., Lim, S.-Y., Choi, B.-Y., & Choi, D.-H. (2022). Review of recent automated pothole-detection methods. *Applied Sciences, 12*(11), 5320. https://doi.org/10.3390/app12115320

This review looks at automated pothole detection through vibration sensing, vision-based methods, and three-dimensional reconstruction. It describes the data collection and processing approaches used in previous research and discusses directions for future development. For RoadWatch, it provides a basis for comparing possible sensing approaches and justifying our initial camera based design. It will also help our team identify limitations and determine which improvements are feasible within the project's scope.

Credible Web Resources

Miller, J. S., & Bellinger, W. Y. (2014). *Distress identification manual for the Long-Term Pavement Performance Program* (5th revised ed., Report No. FHWA-HRT-13-092). Federal Highway Administration. https://www.fhwa.dot.gov/publications/research/infrastructure/pavements/ltpp/13092/

This federal resource provides definitions, measurement procedures, and severity guidance for pavement distress, including potholes and cracks. It establishes consistent terminology for

collecting and describing road condition information. RoadWatch can use this guidance to develop clear damage categories and consistent records. It also provides a reference for exploring severity assessment while recognizing that some measurements require information beyond ordinary camera images.

City of Reno. (n.d.). *Public Works*. Retrieved October 16, 2026, from https://www.reno.gov/government/departments/public-works/

The City of Reno's Public Works website explains local infrastructure responsibilities and directs residents to Reno Direct for reporting potholes and other street concerns. It also links to pavement condition and roadway maintenance jurisdiction maps. This official local resource helps RoadWatch understand the existing reporting environment and relevant geographic information. It will inform our requirements for mapping detections and presenting useful information to municipal staff.

10. Contributions of Team Members

Team Member	Activities Completed	Hours
Christopher Thrasher	Formatted document; wrote table of contents, sections 2, 3.1, 3.2, 4.1, 4.2, & 4.3	7
Vansh Schdeva	Sections 6, 7, 8, part of section 9	5.5
Victor Rosales	Sections 3.1, 3.2, 9	4
Noe Rios	Sections 5.1, 5.2, 5.3	5